



Deep learning-based stress detection using encoded physiological signal representations

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Abstract

The increasing occurrence of stress-related health issues has highlighted the need for accurate and reliable stress detection systems. This study proposes a machine learning-based framework for stress detection using two widely recognized multi-modal physiological datasets: NEURO and WESAD. We apply both classical machine learning models and deep learning architectures to classify different types of stress. To effectively capture the temporal and structural patterns in physiological time-series data, the raw signals are converted into image representations using Gramian Angular Summation Fields (GAFs), Gramian Angular Difference Fields (GAFd), and Markov Transition Fields (MTF). While conventional methods transform the data along the time axis, our approach performs transformations along the feature axis. This enables the model to capture feature-centric patterns that are particularly relevant for distinguishing stress states. The classification performance is enhanced by the Convolutional Neural Network (CNN) through the acquisition of spatial features from these visual representations. The Gradient-weighted Class Activation Mapping (Grad-CAM) visualizations highlight the most influential input regions that drive the model's decision. This, in turn, offers insights into the physiological indicators of stress. Consistent strong generalization is observed across both datasets, with feature-axis transformations based deep learning models achieving around 98.3% accuracy, and classical models like XGBoost and LightGBM achieving approximately 99% accuracy. The integration of Grad-CAM validates the relevance of learned features strengthening the model's interpretability. Altogether, our approach clearly demonstrates the strong potential of integrating advanced signal transformation techniques with explainable deep learning to create robust and interpretable stress detection systems.

Keywords Explainable AI (XAI) · Mental Health Monitoring · Physiological Signals · Stress Detection · Time-Series Encoding

1 Introduction

Stress is a nonspecific physiological response to demands which may be internal or external demands that has a strong effect on the physical health, psychological functioning, and behavioral consequences of individuals. Over 70% of individuals in the United States report facing stress in their daily lives (The American Institute of Stress 2020). Persistent exposure to stress has been associated with a compromised immune response (Segerstrom and Miller 2004) and a heightened likelihood of developing conditions such as cancer (Cohen et al. 2007), heart disease (Cohen et al. 2007; Steptoe and Kivimaki 2013), depression and psychological disorders (Van Praag 2004), diabetes (Heraclides 2012; Mitra 2008) and substance dependence (Al'Absi 2006). Although moderate stress is adaptive, chronic exposure is responsible for an increased risk for disorders such

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as insomnia, obesity, cardiovascular disease, and specific forms of cancer. Furthermore, chronic stress had always been strongly linked with mental disorders like depression and anxiety.

In the modern era, chronic stress has become ubiquitous due to pressures of contemporary life. Stress had caused 50% of total work-related sickness in 2020/21–13% more than 2015/16, as indicated by the British Health and Safety Executive. Similarly, 80% of workers are currently suffering from day-to-day stress as per the American Institute of Stress, and most require intervention to manage it. These statistics underscore the urgent need for effective, real time stress detection that can support immediate intervention and the avoidance of future health impact.

Traditional methods of stress detection rely primarily on psychological self-report measures, such as the Perceived Stress Scale. These measures, however, are inherently subjective and capture merely a fleeting state of an individual and are subject to personal bias and social desirability. Therefore, automatic, objective, and non-invasive stress detection methods have gained significant attention.

The continuous tracking of physiological signals, including blood volume pulse (BVP), electrodermal activity (EDA), heart rate (HR), heart rate variability (HRV), body temperature, and movement via accelerometers has been possible due to the widespread adoption of Internet of Things (IoT) and wearable devices. The dynamic physiological responses captured enable real-time assessment of an individual's condition and play a crucial role in stress detection studies.

The advances in Machine Learning (ML) techniques has substantially elevated the operational capacity of automated stress recognition frameworks. Manual feature design utilised by ML approaches is replaced by the direct extraction of meaningful and scalable representations from raw input signals in Deep Learning (DL) architectures. The robustness and generalisation ability of stress prediction frameworks are significantly enhanced by the incorporation of diverse physiological measurements.

A growing body of work converts physiological time-series signals into image-based formats. These include Gramian Angular Fields (GAF), MTFs, recurrence plots, grayscale mappings, spectrograms, and scalograms which in turn enable the learning models to effectively capture complex temporal and spatial properties.

In our study, a CNN-driven framework encodes the physiological signal representations to support reliable, non-invasive stress identification. The integration of image-based signal encoding and deep learning techniques delivers a robust framework for reliable stress detection under real-world conditions.

The feature-axis transformation of physiological data is a novel preprocessing strategy employed by our work to enhance deep learning-based classification. Our method differs from previous stress detection models that transformed data on time-axis (e.g., STREDWES), which only captures temporal properties of single-sensor streams. Instead, we encode data based on feature-axis transformation, allowing the model to simultaneously learn relationships across multiple modalities (EDA, TEMP, BVP, HR) concurrently during a defined time period. The combined representations using feature-axis encoding capture inter-feature relationships that better represent physiological patterns of stress.

In addition, we utilize Grad-CAM to help explain model output to indicate stress correlated features that correspond with known biomarkers of sympathetic activation. In summary, this combination of encoding imaging + DL + interpretability is a novel, explainable, and physiologically-grounded approach to stress detection.

2 Literature survey

Researchers have invented numerous methods to interpret physiological signals collected by body-worn sensors for the purpose of stress detection and emotion classification (Sioni and Chittaro 2015). Previous studies have demonstrated that physiological signal analysis can consistently detect human stress (Lundberg et al. 1994).

Extensive research has focused on utilizing physiological signals for the identification of stress (De Santos Sierra et al. 2011; Smets et al. 2018; Wang et al. 2013). Most earlier approaches looked at a combination of physiological signals, such as those recorded by electrocardiogram (ECG) sensors (Kyriakou et al. 2019), EDA sensors (Karthikeyan et al. 2013), and electromyography (EMG) sensors (Setz et al. 2010), to detect stress and related states.

STREDWES (Quadrini et al. 2024) is a stress detection framework that transforms physiological signals into image representations, which are then classified using CNNs. The study evaluates different encoding methods, such as GAFs, GAFd and MTF, using datasets like NEURO, SWELL, and WESAD. Results demonstrate that STREDWES outperforms traditional methods, offering a robust, non-invasive approach to stress detection through wearable sensor data.

The authors in Li and Liu (2020) introduced two deep learning models—a one-dimensional convolutional neural network (1D-CNN) and a multilayer perceptron (MLP)—for the classification of stress and emotions from physiological signals. They were trained end-to-end from raw sensor-captured chest- and wrist-worn signals, without manual feature engineering. Such high accuracies of 99.80% binary stress detection and 99.55% three-class emotion classification are

demonstrated in the study and vastly outperform conventional machine learning methods. It highlights the capability of deep learning for long-term, non-intrusive stress and emotion monitoring.

Tzirakis et al. (2017) introduced a deep neural network capable of processing video segments in order to identify stress. Mirsamadi et al. (2017) used a recurrent neural network for detecting stress via speech processing in another study.

Rastgoo et al. (2019) introduce a multimodal deep learning approach for detecting drivers' stress levels. It combines ECG signals, vehicle dynamics, such as steering and braking behaviors, and contextual information, such as weather and road conditions. CNNs and LSTMs together detect complex temporal and spatial patterns across modalities. Experimental findings revealed that this approach outperforms traditional machine learning models in terms of classification accuracy.

The authors in Nandan et al. (2024) categorize emotional states from EEG signals transformed to spectrograms. They use a number of techniques, such as Random Forest, K-Nearest Neighbors (KNN), and deep learning models like LSTM and CNNs. According to their results, CNNs from spectrograms perform better than typical sequence models for enhancing emotion classification of six emotions—fear, anger, joy, sadness, surprise, and disgust.

de Souza et al. (2022) used a pipeline for stress classification using physiological data from wearable devices and a Recurrent Neural Network (RNN) model. The pipeline uses various pre-processing techniques like Fourier analysis, sliding window labeling, and rolling z-score normalization. The RNN model in the WESAD dataset achieves 86% accuracy for multi-class classification and 96.5% accuracy for binary stress detection, outperforming classical machine learning strategies.

The paper “Review on Psychological Stress Detection Using Biosignals” (Giannakakis et al. 2022), offers an in-depth investigation into psychological stress detection using different biosignals. The abstract identifies the physiological and physical biosignals (EEG, ECG, EDA, EMG, speech, respiration, skin temperature, pupil size, eye activity) that react to stressors and stresses their involuntariness and reliability. The research seeks to determine reliable, efficient, and reproducible biosignal features for effective stress detection. The procedure takes the form of a survey of literature for signal-specific studies under conditions of stress, classification of features according to type and statistical significance. The effort takes the form of detailed comparison of signal patterns, multimodal analysis, and assessment of stress induction protocols. Findings indicate stable changes in signals such as EEG asymmetry, heart rate acceleration, and skin conductance levels. The evaluation

leads to applied recommendations and insights into the stability of biosignal-based stress indices for psychophysiological research and real-time use.

The article “Monitoring Stress with a Wrist Device Using Context” (Gjoreski et al. 2017) offers a new machine-learning-based approach to real-time psychological stress detection from a wearable wrist device. The article starts with the problem of stress detection in real-life based on its subjectiveness and physiological nature. The suggested method consists of a three-module system: lab-trained stress detector, activity recognizer, and context-based classifier using biosignal data (BVP, EDA, HR, ST, IBI, ACC) and context inputs (activity level, time, etc.). The experiments were performed on 21 subjects under lab settings and were further applied to 55 days of real-life data from 5 users. Results indicated the context-aware approach outperformed the lab-only model substantially, with 70% recall and 95% precision in identifying stress events. The study verifies the robustness and feasibility of the system for real-time, unobtrusive monitoring of stress.

The study “DeStress: Deep Learning for Unsupervised Identification of Mental Stress in Firefighters from Heart-rate Variability (HRV) Data” (Oskooei et al. 2019) introduces an unsupervised deep learning model designed to detect mental stress in firefighters using raw RR interval data. The abstract describes the application of conventional K-Means with handcrafted features, convolutional autoencoders (CAE), and LSTM autoencoders (LAE) along with DBSCAN and KNN classifiers. The process involves HRV data collection from 100 trainee firefighters during stress-inducing training, data preprocessing, feature engineering, and unsupervised model training. The study demonstrated that K-Means was unable to retrieve useful clusters, whereas CAE performed better than LAE in identifying unique, stress-related patterns validated using HRV markers like RMSSD and LF/HF ratio. Analysis validated CAE's clusters in line with physiological stress indicators, illustrating the model's capability for real-world, label-free stress sensing in hazardous professions.

Later research in stress detection has more and more utilized multimodal physiological signals and sophisticated model structures to enhance accuracy as well as interpretability. Wang et al. proposed *TAGformer*, a new deep learning system based on topology-adaptive graph convolution and transformer encoders for integrating features from multiple physiological modalities, such as ECG, EMG, RESP, and skin temperature, to recognize pilot stress precisely. Their model exhibited better performance in classifying more than one stress level by directly modeling inter-signal relations, and their performance on a two-class task was over 93% (Wang et al. 2024). Likewise, in Al-Naji et al.'s research, hand-crafted features from biosensor data from

wearable devices were maximized with the help of classic machine learning classifiers augmented with explainable AI methods such as SHAP and LIME. This paper focused on interpretable stress classification and emphasized the value of feature importance analysis to clinical understanding (Shikha et al. 2024). These findings are in-line with our interest in explainable deep models and confirm the increasing importance of transformer-based fusion and interpretable physiological modeling in stress analysis.

3 Dataset description

For performance assessment, our framework is tested on two openly accessible multimodal physiological datasets, NEURO and WESAD. Each dataset provides measurements of human physiological signals under carefully controlled conditions intended to elicit stress and relaxation responses. The datasets include measurements from sensors tracking heart rate, electrodermal responses, and motion, recorded across various controlled experimental environments.

3.1 NEURO dataset

The development and evaluation of several stress detection approaches leveraged the NEURO dataset (Birjandtalab et al. 2016). A key feature of the NEURO dataset is its structured stress-induction design, which alternates physical, cognitive, and emotional challenges with relaxation periods to facilitate baseline comparisons.

The NEURO data set, acquired from 20 participants, was obtained with institutional review board ethics approval from the University of Texas at Dallas Institutional Review Board (UTD IRB #12–29). The experiment design includes seven consecutive trials that are intended to elicit and observe physiological change under different types of stress:

1. **First Relaxation (5 minutes):** Initial baseline in a relaxed, calm state.
2. **Physical Stress (5 minutes):** The protocol includes standing for one minute, followed by two minutes of paced treadmill walking (1 mph) and two minutes of brisk walking or running (3 mph).
3. **Second Relaxation (5 minutes):** Recovery period to monitor recovery toward baseline following physical activity.
4. **Mini-Emotional Stress (40 seconds):** Spontaneously elicited during reading of cognitive task instructions, which unexpectedly provoked stress responses from multiple subjects.
5. **Cognitive Stress (5 minutes):** A high-order, mathematically demanding exercise on a working memory item

(counting down from 2485 in sevens for three minutes) and then the Stroop task (2 minutes) with auditory error feedback in the form of a buzzer.

6. **Third Relaxation (5 minutes):** Assessment of recovery during relaxation following cognitive load.
7. **Emotional Stress (5 minutes):** Comprised of a one-minute anticipation phase after viewing a suspenseful clip from the horror movie *The Horde* (4 minutes).
8. **Fourth Relaxation (5 minutes):** Final recovery period to wrap up the session.

By capturing dynamic physiological responses to various stressors, the NEURO dataset serves as an ideal resource for machine learning-driven stress detection and assessment.

3.2 WESAD dataset

As a contribution to affective computing, the WESAD dataset (Schmidt et al. 2018) provides valuable data for identifying emotional and physiological responses. It evaluates the application of chest- and wrist-worn sensors to detect stress and other emotional responses from the gathered physiological data.

Information from 15 individuals, including multimodal motion and physiological signals, was recorded as part of a controlled laboratory study. Two wearable devices, the RespiBAN chest sensor and Empatica E4 wrist sensor, recorded the data. The sensors recorded various signals such as ECG, EDA, EMG, breathing, body temperature along with tri-axial acceleration. Both of the data from the devices were synchronized by a double-tap gesture, with reference being drawn from the RespiBAN device.

RespiBAN device, worn around the chest, records a number of signals like ECG, EDA, EMG, breathing, body temperature, and acceleration in three axes with sampling frequency at 700 Hz. Empatica E4, worn around the wrist, records blood volume pulse (BVP), EDA, body temperature, and acceleration in three axes. Empatica E4 samples BVP and EDA at 64 Hz and body temperature at 4 Hz.

The SX.pkl files contain synchronized data from both the RespiBAN and Empatica E4 devices. These files are structured as dictionaries and include:

- **subject:** The unique participant identifier.
- **signal:** A dictionary containing raw data from both devices:
 - **chest:** Data from the RespiBAN device (ACC, ECG, EDA, EMG, RESP, TEMP).
 - **Wrist:** The Empatica E4 wearable collected motion, cardiovascular, electrodermal, and temperature data through ACC, BVP, EDA, and TEMP sensors.

- **Label:** The experimental protocol supplied condition annotations, sampled at 700 Hz.

The study's classification task is based on four categories specified by the protocol: **Baseline**, **Stress**, **Amusement**, and **Meditation**. Labels 0, 5, 6, and 7 are excluded from the task.

4 Methodology

In this section, we present the methodology employed to detect stress from the NEURO and WESAD datasets. The complete workflow is illustrated in Fig. 1. Raw physiological signals from both NEURO and WESAD datasets undergo normalization and segmentation using sliding windows. These are transformed into two-dimensional encodings (GAF, GAFd, MTF) for CNN-based classification. Classical ML and deep learning models are applied to NEURO, while Grad-CAM visualization is used only for WESAD to interpret physiologically relevant stress patterns. The fusion of these techniques forms the foundation of our stress detection framework.

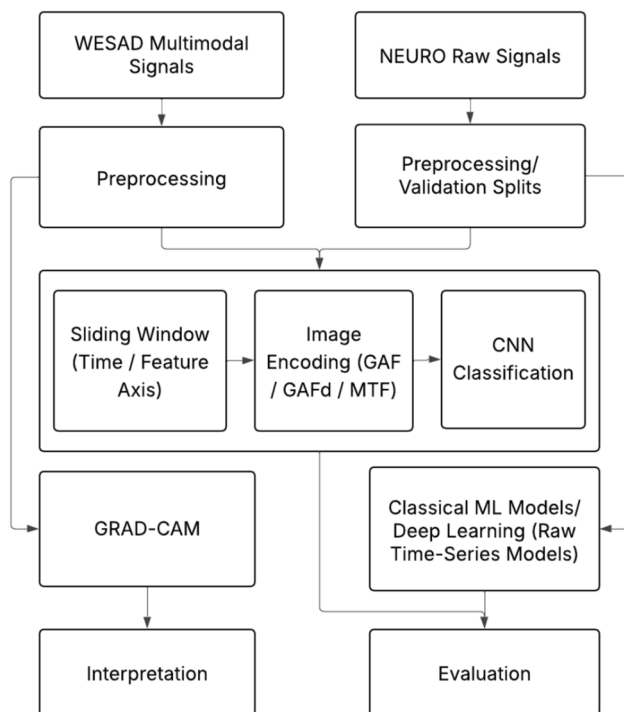


Fig. 1 Overview of the proposed stress detection framework. NEURO and WESAD signals undergo preprocessing, sliding-window segmentation, and image encoding (GAFs, GAFd, MTF) for CNN-based classification. Grad-CAM is applied only to WESAD for interpretability

4.1 NEURO

Training and evaluation using the NEURO dataset have been carried out for various models of stress detection. The processes employed in the current work include data split approaches and utilizing the conventional machine learning models, deep learning of raw time-series signals, as well as image-based deep learning methods.

4.1.1 Data splitting techniques

To allow robust testing of the models, two methods of data splitting were used:

- **Leave-One-Subject-Out Validation (LOSO):** Here, data from one subject was not included in the dataset, and training on the data of all other subjects was done. Then, the model was tested on the data of the subject that was left out. This was done for each subject so that the model was tested on unseen data. LOSO validation provided the model with information on how well it could generalize over different individuals, giving a more reliable measure of how well it was performing.
- **Cross-Validation Per Subject:** For this method, a unique model was trained for each single subject. For each subject, data was divided 80/20 training/test. This allowed model performance to be tested per subject, which gave insight into model generalizability to new data from the same individual. Accuracy was calculated per subject's model to establish performance.

4.1.2 Classical ML models

In this subsection, we applied various classical machine learning models to the NEURO dataset for stress detection. The models evaluated include Support Vector Machines (SVM) with Radial Basis Function (RBF) and Polynomial (Poly) kernels, CatBoost, XGBoost, LightGBM, Random Forest, Multi-Layer Perceptron (MLP), K-Nearest Neighbors (KNN), Decision Tree, Naive Bayes, Fisher's Linear Discriminant Analysis (FLDA), Least Squares (LS), and a Voting Ensemble of LightGBM, CatBoost, and Random Forest classifiers.

Features such as accelerometer axes (ax , ay , az), Electrodermal Activity (EDA), temperature (t_{temp}), oxygen saturation ($spo2$), and heart rate (hr) were extracted from the NEURO dataset and fed as input to each of the above models. Each row of the dataset served as an independent data instance during model training. Each data instance was obtained after assembling fixed-duration physiological signal segments. Specifically, for each subject and each stress category (Relax, PhysicalStress,

EmotionalStress, CognitiveStress), the signals were split into 8 samples in the case of ax, ay, az, EDA, and temp, and 1 sample in the case of spo2 and hr. These fragments were finally glued together in order to form a single feature vector for representing one example.

This systematic approach ensured that all models were supplied with uniform and well-formatted input data, enabling fair and stable comparison of performance.

4.1.3 Deep learning on raw time-series

In this, we employed deep learning models to analyze raw time-series data from the NEURO dataset for stress detection. The models being considered are CNNs, Long Short-Term Memory networks (LSTMs), and a CNN-LSTM hybrid architecture.

Data Preparation The initial time-series data had features such as ax, ay, az, temp, EDA, hr, and spo2. They were scaled through standard scaling in order to achieve uniformity of features.

Sliding Window Approach To make the data ready for deep learning models for LOSO approach, we used a sliding window technique. Each window had a fixed number of time steps (e.g., 100) with a particular step size (e.g., 20). For every window, the target class was the most frequent label. This technique helped us transform the time-series data into a suitable format for training sequential models.

Data Input for Models The formatted data was input into each of the deep learning models in the following way:

- Each sample (or window) of data was provided as a 2D array, with the rows as the time steps in the window and columns as the different features.
- These 2D arrays were stacked together to form a 3D array, the first dimension referring to the number of windows, the second dimension referring to the time steps, and the third dimension referring to the features.
- This 3D array was used as input to the CNN and LSTM models so that the models can process the temporal and spatial information in the time-series data.

This formal process ensured that all models were supplied with consistent and well-defined input data so that good learning and prediction of the stress levels from the raw time-series data could occur.

Model Architectures CNN Model: The CNN architecture was designed to extract spatial features from time-series data. It consisted of Conv1D layers with ReLU activation to extract local patterns, followed by MaxPooling1D layers to down-sample, and dense layers to perform the final classification.

LSTM Model: It focuses on learning time-series signal temporal dependencies. The model used LSTM layers to represent information sequentially well and dense layers for making class predictions.

CNN-LSTM Model: It is a combined model that availed the strength of CNN as well as LSTM. Spatial features were extracted through initial Conv1D and MaxPooling1D layers, and the extracted features were passed through LSTM layers for temporal pattern extraction. Final dense layers did the classification.

4.1.4 Time-series to image-based deep learning

In this sub-section, we had used deep learning models on the NEURO dataset by converting the time-series data into representations based on images. This was done using techniques like GAFs, GAFd and MTF in order to transform the time-series data into 2D images, which were fed as input to CNNs.

Data Preparation Raw time-series data had the following attributes like ax, ay, az, temp, EDA, hr, and spo2. Features were standardized employing standard scaling in order to evade any variance over different features.

Sliding Window Approach Two types of Sliding Window Approaches were used in for this task namely:

Sliding Window Approach Along Time Axis: To prepare the data for conversion into images, we employed a sliding window approach. All window consisted of a fixed number of time steps (e.g., 100) with a specified step size (e.g., 20). For each window, the most frequent label was assigned as the target class. This method enabled us to transform the time-series data into a format appropriate for converting into images.

Sliding Window Approach Along Feature Axis: The data is initially normalized using MinMaxScaler (−1 to 1), segmented with a sliding window, and reshaped by feature. Each window is transformed into 2D images using GAFs, GAFd and MTF to capture temporal and inter-feature patterns. Class imbalance is addressed via equal sampling across labels (1–4) Fig. 2.

Time-Series to Image Conversion Window-based encoding of the physiological signals into visual representations was performed using the following methods:

Gramian Angular Fields (GAFs): Time-series signals are encoded into a Gramian matrix, preserving the angular relationships between temporal samples.

Gramian Angular Fields with differences (GAFd): While akin to Gramian Angular Fields, it encodes the sequential differences between neighboring time points.

Markov Transition Fields (MTFs): Creates a Markov transition matrix representing the probability of transitioning between time points.

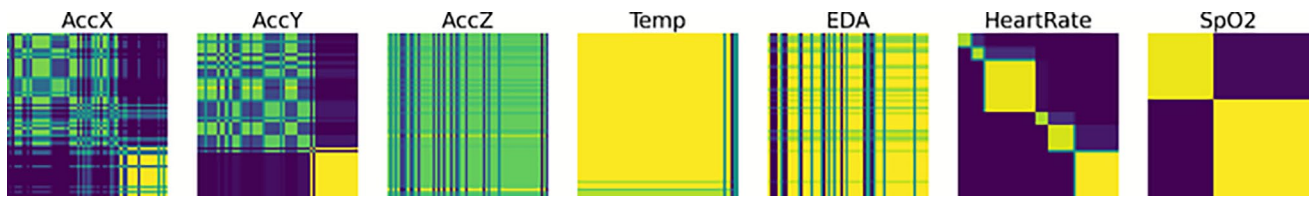


Fig. 2 Example GAFd images for each sensor modality from a single window of the NEURO dataset, labeled as Physical Stress

Data Input for Models The converted image data was passed into the CNN models as 4D arrays, where dimensions corresponded to the number of windows, image height, image width, and the number of channels (features). Since we are using 7 features the input dimension to the model will be $64 \times 64 \times 7$ as shown in image 2 (This visualization is intended solely to illustrate the transformation of normalized physiological time-series segments into pictorial GAFd representations).

CNN Model Architecture The CNN model architecture consisted of:

- **Input Layer:** Accepted images of shape determined by the conversion technique.
- **Convolutional Layers:** Three layers, each with 32, 64, and 128 filters, respectively, employing 3×3 kernels and ReLU activation.
- **MaxPooling Layers:** Applied after each convolutional layer with a 2×2 pool size.
- **Flatten Layer:** Converted 2D feature maps into a 1D feature vector.
- **Dense Layers:** Consisted of two fully connected layers with 128 and 64 units, respectively, utilizing ReLU activation, followed by a dropout layer for regularization.
- **Output Layer:** A dense layer with softmax activation for class probability distribution.

This architecture enabled the CNN models to effectively learn and predict stress levels from the spatial patterns and temporal dynamics captured in the image representations.

4.2 WESAD dataset

4.2.1 Data preparation

Sliding Windows along the Time as Axis We transform the physiological time-series data of the WESAD dataset into image representations of raw sensor signals by applying a sliding window approach, which is well-suited to deep learning models. The data from each subject's.pkl file is loaded in order to obtain multimodal chest sensor data (e.g., ACC, ECG, EDA, EMG, Resp, Temp) and then join it with its corresponding ground truth labels. Segments labeled

from classes 1 to 4 (for various affective states) only are retained for processing.

The sliding window technique employs a constant window size of 294 time steps with a stride of 147, providing 50% overlap between adjacent windows for adequate temporal coverage. Windows are created for every physiological feature to identify short-term temporal dependencies. Here we are treating each feature independently and the windows are being constrained for each feature across time along multiple feature. For the purpose of minimizing computational load and storage, only 25% of the windows are randomly sampled.

The windowed data is normalized between -1 and 1 using MinMaxScaler, ensuring compatibility with the GAF transformation. GAF, implemented via `pyts.image`. GramianAngularField with the 'summation' method, converts each normalized 1D time-series window into a 2D image of size 105×105 , encoding temporal correlations as angular representations. These images are saved as .png files in directories organized by subject and label class.

Furthermore, feature extraction was enhanced using methods such as GAFs and MTFs. Temporal descriptors were extracted from GAF images, while MTFs simultaneously capture the time and frequency domains of multivariate physiological signals. This rich set of features facilitates model training that increases the classification accuracy in emotion prediction tasks.

Sliding Windows along the Features as Axis

Input to this approach comprises of time-series signals like ECG, EDA, and EMG across numerous participants obtained from the WESAD dataset. Initially, the data undergo loading and normalization before being arranged into a structured format. Significant features include applying sliding windows to the dataset, which is reshaped along the feature dimension to analyze individual segments.

Each window, being part of time-series, undergoes GAF transformation and converts data to 2D images. Such images provide information regarding temporal as well as feature correlation. The MinMaxScaler normalizes time-series data in a consistent fashion such that every feature remains within -1 and 1 .

Data is balanced by sampling an equal number of instances per label (1, 2, 3, 4). Transformed GAF images are saved in a structured directory by subject and label to

provide organized access for additional analysis or machine learning tasks.

Here we are treating multiple features together across time for windows generation.

4.2.2 Deep learning approach

The model employs a CNN with PyTorch for image classification. The dataset contains images labeled into four labels. A 'GAFImageDataset' class is used to load and preprocess images, resizing them to 32x32 pixels and converting them to tensors. The CNN model ('CustomCNN') is made up of three convolutional layers with ReLU activation functions, followed by max-pooling and adaptive average pooling, culminating in a fully connected layer for classification.

Adam optimizer and cross-entropy loss are used to train the model, with metrics being Precision, Recall, F1-score, and Accuracy, computed via PyTorch's 'torchmetrics'. We use an 80/20 train/test split on the dataset, and the training loop is done with tqdm for tracking progress. The model is trained on a GPU if available, and the results are printed for each epoch. The model employs a learning rate of 0.001, batch size of 32, and trains for 10 epochs with the performance metrics calculated based on macro-average over 4 classes.

4.2.3 Grad-Cam

To visualize the key regions of the image that influenced the CNN model's predictions, Gradient-weighted Class Activation Mapping (Grad-CAM) was used to highlight the areas that played the most significant role in the model's decision-making process. It generates these heatmaps by calculating the gradients of the target class with respect to the last convolutional layer. The detected influential regions are then projected onto the GAF or MTF matrices to provide a structured spatial representation. Window-based mapping of the generated heatmaps and the GAF/MTF matrices promotes enhanced analysis and gains interpretable insights into the model's decision-making. This methodology, therefore, enables an in-depth comprehension of the CNN's predictive reasoning.

4.3 Evaluation metrics

To evaluate the effectiveness of the proposed classification methods, a set of diverse performance indicators obtained from the confusion matrix (Table 1) was used. The overall

Table 1 Confusion matrix

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

predictive outcomes of a binary classification model are depicted using a confusion matrix.

Various aspects of model performance are examined using the metrics outlined below:

4.4 Accuracy

The accuracy measures the proportion of predictions that match the true outcomes compared to the total number of instances.

$$\text{Accuracy} = \frac{TP + TN}{FN + TP + FP + TN} \quad (1)$$

4.4.1 Precision (positive predictive value)

This metric evaluates the validity of positive classifications by comparing them with actual positive cases.

$$\text{Precision} = \frac{TP}{FP + TP} \quad (2)$$

4.4.2 Recall (true positive rate)

This metric evaluates the coverage of positive instances, demonstrating the count of true positive predictions.

$$\text{Recall} = \frac{TP}{FN + TP} \quad (3)$$

4.4.3 F1 score

A unified performance measure that combines recall and precision into an inverse weighted mean, particularly beneficial for imbalanced datasets.

$$\text{F1 Score} = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \quad (4)$$

By incorporating the relationships from Equations (2) and (3), the expression can be equivalently written as:

$$\text{F1 Score} = \frac{2TP}{2TP + FN + FP} \quad (5)$$

4.4.4 Specificity (true negative rate)

This metric shows how many true negatives are identified from all actual negative instances.

$$\text{Specificity} = \frac{TN}{FP + TN} \quad (6)$$

Table 2 Summary of model parameters and hyperparameters

Parameter	NEURO Dataset	WESAD Dataset
Window Size	100	294
Stride	20	147
Normalization Range	[-1, 1] MinMax	[-1, 1] MinMax
Encoding	GAF, GAFd, MTF	GAF/GAFd
CNN Layers	3 Conv + 2 FC	3 Conv + 1 FC
Feature Axis Used	Yes	Yes
Optimizer	Adam (lr = 0.001)	Adam (lr = 0.001)
Epochs	10	10
Accuracy (CV)	98.3%	81.5%
Accuracy (LOSO)	79.9%	–

When considered collectively, these indicators provide a balanced evaluation of the model's classification performance, especially under conditions of class imbalance.

4.5 Summary of model parameters and hyperparameters

The experimental parameters, including data preparation, model setup, and hyperparameter choices, are presented in Table 2, providing a transparent methodological process.

We employed ReLU as our activation function, dropout rate of 0.3 after each dense layer, batch size of 32, and softmax output for four-class classification. In each LOSO split, the testing subject is completely isolated from all training and validation samples to avoid data leakage in any form.

5 Results

This section presents the experimental results on the NEURO and WESAD datasets. A detailed comparative analysis of the model performance and their corresponding Grad-CAM heatmap insights are also provided.

5.1 NEURO dataset

Cross-Validation per Subject and Leave-One-Subject-Out (LOSO) were the two experimental frameworks used to assess the model performance on the NEURO dataset using the metrics discussed in section 4.3. The dataset is composed of four target classes reflecting different stress conditions: Cognitive, Physical, Emotional, and Relaxation.

5.1.1 Evaluation of sliding window strategies

For the GAF, GAFd, and MTF transformations, we considered sliding windows across temporal and feature dimensions to evaluate the efficacy of each of them. Cross-validation results showed that the feature-axis method excelled the time-axis approach consistently across all encodings and

Table 3 Performance of MTF encoding across sliding window strategies and validation schemes

Metric	CV (Time)	CV (Feature)	LOSO (Time)	LOSO (Feature)
Accuracy	72.61	98.27	79.97	52.28
F1-Score	70.95	98.26	79.23	35.90
Precision	72.47	98.28	80.39	52.27
Recall	72.61	98.27	79.95	52.27
Specificity	82.16	99.12	88.77	47.73

Table 4 Performance of GAFs Encoding across Sliding Window Strategies and Validation Schemes

Metric	CV (Time)	CV (Feature)	LOSO (Time)	LOSO (Feature)
Accuracy	64.10	98.36	69.96	51.10
F1-Score	61.80	98.35	68.94	46.78
Precision	61.12	98.36	70.34	47.39
Recall	64.10	98.35	69.98	51.11
Specificity	78.47	99.20	84.37	70.83

Table 5 Performance of GAFd Encoding across Sliding Window Strategies and Validation Schemes

Metric	CV (Time)	CV (Feature)	LOSO (Time)	LOSO (Feature)
Accuracy	67.50	98.35	77.89	51.19
F1-Score	65.52	98.35	76.77	50.40
Precision	67.78	98.36	78.63	55.26
Recall	67.50	98.35	77.85	51.89
Specificity	79.04	99.21	86.45	76.99

achieved higher F1-scores (98.3%), as detailed in Tables 3, 4, 5. In LOSO experiments, the time-axis strategy for MTF outperformed the feature-axis approach with an F1-score of 79.23%, while the latter reached only 35.9%. This illustrates that feature-axis windows benefit subject-specific models, and time-axis windows improve generalization across subjects.

Overall, cross-validation yields markedly better results than LOSO for all models and feature encodings. As a specific example, the combination of MTF and feature-axis sliding in cross-validation achieved an F1-score of 98.26% (Table 3), while LOSO's top-performing configuration with MTF and time-axis sliding only reached 79.23%. These results indicate the significant role of subject-to-subject variability.

Tables 3, 4, 5 demonstrate significant performance variations between encoding strategies. The persistently higher recall and F1-scores of LOSO+MTF implies improved generalization of stress-related temporal dynamics across subjects by explicitly capturing the state transition probabilities. The decline in performance of LOSO+GAF and LOSO+GAFd feature-axis approaches indicates the heavy dependence on individual baseline characteristics in the absence of temporal structures. The discrimination is

non-stress conditions is easily attained in comparison to the stress-related classes as reflected by the high specificity across encodings under CV. Practically, feature-axis GAF excels for personalized stress monitoring, while time-axis MTF is preferable for subject-independent recognition.

5.1.2 Evaluation of models using LOSO and CV

The MTF-based CNN achieves top performance under LOSO, with F1-score 79.23% and accuracy 79.97% (Table 6), demonstrating strong generalization. The improvement arises from MTF's ability to encode state transitions, making it more robust to subject-specific physiological variations than static features. Ensemble models like Random Forest, XGBoost, and CatBoost perform competitively, indicating that tree-based aggregation partially addresses cross-subject variability. LSTM and CNN+LSTM achieve lower accuracy, reflecting the difficulty of learning cross-subject temporal dependencies. MTF achieves the highest recall (79.95%) and maintains high specificity, improving stress detection accuracy and reducing false positives in real-world scenarios.

The class-wise predictions of the GAFd model using feature-axis transformations under the CV setup is shown in the confusion matrix of Fig. 3. The class-wise performance was found to be highest for the Relax class (89.51%) followed by the Physical Stress class (77.92%), showcasing their unique physiological markers. Moderate accuracy of

Table 6 Performance of models using Leave-One-Subject-Out evaluation on the NEURO dataset

Model	Accuracy	Precision	Recall	F1-score	Specificity
MLP	64.32	57.67	51.08	48.23	64.32
KNN	71.62	58.63	50.85	49.11	91.40
Random Forest	77.63	65.23	59.92	58.85	93.35
Decision Tree	77.19	49.03	50.20	47.98	93.56
Naive Bayes	72.53	60.09	48.43	49.26	92.11
FLDA	77.24	55.41	52.47	49.88	87.59
LS	41.71	25.81	30.30	20.83	82.21
XGBoost	77.10	65.62	59.48	58.14	93.28
LightGBM	73.36	61.94	56.55	53.76	91.79
CatBoost	76.09	65.46	59.79	57.89	92.68
SVM (RBF)	71.07	61.06	53.32	51.55	90.91
SVM (Poly)	72.72	63.07	49.91	47.95	91.66
Voting Ensemble	75.73	64.53	58.85	56.75	92.59
1D CNN	61.36	63.78	61.36	58.51	88.53
LSTM	60.64	63.80	60.64	57.62	88.62
CNN + LSTM	58.91	63.87	58.91	56.04	88.77
GAFs	69.96	70.34	69.98	68.94	84.37
GAFd	77.89	78.63	77.85	76.77	86.45
MTF	79.97	80.39	79.95	79.23	88.77

60.94% and 56.90% is observed for the Cognitive and Emotional Stress classes due to misclassification as Relax. This is the consequence of overlapping autonomic responses in lower-intensity and recovery phases.

The availability of subject-specific training data with reduced inter-subject variability results in high cross-validation performance as shown in Table 7. The combined

Fig. 3 Confusion matrix for the GAFd feature-axis model under cross-validation (CV) on the NEURO dataset, illustrating class-wise prediction accuracy across Relax, Physical, Cognitive, and Emotional stress states

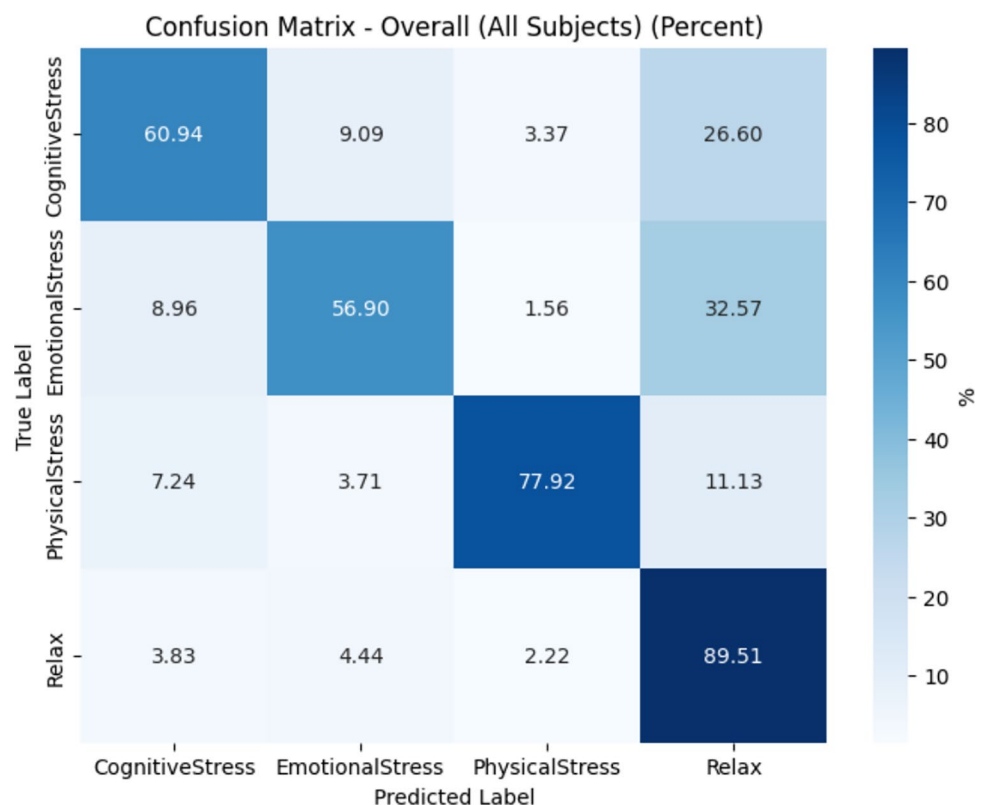


Table 7 Performance of models using Cross-Validation Per Subject on the NEURO dataset

Model	Accuracy	Precision	Recall	F1-score	Specificity
MLP	98.84	98.86	98.84	98.82	99.50
KNN	93.68	76.50	73.75	72.67	98.36
Random Forest	99.26	98.53	97.45	97.91	99.82
Decision Tree	95.64	87.99	86.17	86.20	98.89
Naive Bayes	91.76	86.19	88.06	85.77	97.80
FLDA	91.89	85.55	77.50	79.18	96.10
LS	64.34	40.59	38.02	35.41	91.15
XGBoost	99.24	98.76	97.00	97.78	99.81
LightGBM	99.34	98.86	97.65	98.19	99.84
CatBoost	99.33	98.89	97.65	98.21	99.83
SVM (RBF)	96.72	93.81	88.31	90.05	99.17
SVM (Poly)	95.64	95.35	84.16	86.89	98.90
Voting Ensemble	99.47	99.14	97.96	98.48	99.87
1D CNN	98.24	98.31	98.24	98.23	97.30
LSTM	95.51	95.70	95.51	95.27	96.80
CNN + LSTM	97.98	98.08	97.98	97.98	97.60
GAFs	98.36	98.36	98.35	98.35	99.20
GAFd	98.35	98.36	98.35	98.35	99.21
MTF	98.27	98.28	98.27	98.26	99.12

decision-making capability of the voting ensemble approach was successful at achieving top accuracy (99.47%) and F1-score (98.48%). Tree ensembles, including CatBoost, LightGBM, and XGBoost, perform well by exploiting stable, subject-dependent features. Deep learning models, including 1D CNN and CNN+LSTM, achieve accuracies above 97.9%, indicating that temporal representations are highly effective when individual baselines are present. Notably, image-based encodings (GAF, GAFd, and MTF) maintain uniformly high precision, recall, and specificity (above 98%), suggesting that GAF-based representations provide robust and discriminative features for personalized stress classification. Collectively, these results demonstrate that cross-validation performance represents an upper bound achievable under subject-aware conditions, complementing the more conservative LOSO evaluation.

The significant accuracy gap between Leave-One-Subject-Out (LOSO) and cross-validation (CV) is characteristic of inter-subject physiological variability. CV is close to 98–99% accuracy because training data and testing data share a similar subject's baseline characteristics, whereas LOSO tests on entirely new people with different autonomic baselines, skin conductance levels, and stress reactivity profiles. The LOSO accuracy of 79.97% (MTF) shows moderate generalizability, but the 18-point reduction suggests that models specific to or custom-fit to individuals may be necessary for deployment in practice where variability among individuals dominates.

5.1.2.1 Model comparison with base paper Table 8 presents a direct comparison between the results reported in the original study, which employed Leave-One-Subject-Out (LOSO) validation, and our re-implemented and proposed models evaluated under both LOSO and cross-validation (CV) protocols. Under the same LOSO setting, our implementations consistently outperform the baseline results across all evaluated models, indicating that the observed gains are attributable to improved preprocessing, representation learning, and model optimization rather than differences in evaluation protocol. For instance, a 13% rise in accuracy was seen by the Random Forest model, while CNN+MTF and CNN+GAFd representations showed notable gains in F1-score and specificity. This reflects an improved balance between sensitivity and false-positive control.

All CNN-based models achieve uniformly high performance during the cross-validation evaluation, with accuracy, precision, recall, and specificity exceeding 98%. While CV benefits from subject-specific data and therefore represents an upper bound on achievable performance, these results demonstrate the strong discriminative capacity of the proposed representations under subject-aware conditions. Taken together, the LOSO and CV results highlight the robustness of the proposed framework across both inter-subject and

Table 8 Comparison of Base Paper and Proposed Models (LOSO and CV) on the NEURO Dataset

Model	Setting	Accuracy	F1-score	Precision	Recall	Specificity
Random Forest	Base Paper Quadrini et al. (2024)	64.54%	62.34%	68.73%	64.54%	–
	LOSO (Ours)	77.63%	58.85%	65.23%	59.92%	93.35%
	CV (Ours)	99.26%	97.91%	98.53%	97.45%	99.82%
MTF + CNN	Base Paper Quadrini et al. (2024)	73.34%	72.36%	76.40%	71.27%	83.84%
	LOSO (Ours)	79.97%	79.23%	80.39%	79.95%	88.77%
	CV (Ours)	98.27%	98.26%	98.28%	98.27%	99.12%
GAFs + CNN	Base Paper Quadrini et al. (2024)	71.72%	69.35%	74.94%	67.41%	83.40%
	LOSO (Ours)	69.96%	68.94%	70.34%	69.98%	84.37%
	CV (Ours)	98.36%	98.35%	98.36%	98.35%	99.20%
GAFd + CNN	Base Paper Quadrini et al. (2024)	73.03%	72.49%	76.32%	71.57%	84.14%
	LOSO (Ours)	77.89%	76.77%	78.63%	77.85%	86.45%
	CV (Ours)	98.35%	98.35%	98.36%	98.35%	99.21%

Table 9 Performance of CNN Model with Different Encoding Techniques on the WESAD Dataset - Sliding Windows along the Time as Axis

Model	Accuracy	Precision	Recall	F1-score
GAFs	77.65	71.23	77.65	74.79
GAFd	81.53	92.89	81.53	78.98
MTF	79.02	79.40	79.02	77.49

Table 10 Performance of CNN Model with Different Encoding Techniques on the WESAD Dataset - Sliding Windows along the Feature as Axis

Model	Accuracy	Precision	Recall	F1-score
GAFs	98.48	98.53	98.48	98.47
GAFd	98.92	99.05	98.92	98.97
MTF	99.57	99.52	99.57	99.54

intra-subject evaluation scenarios, supporting its suitability for practical stress recognition applications.

5.2 WESAD dataset

The classification accuracy and overall metric performance of the CNN models as show in Table 9, which were trained on various time series encodings - GAFs, GAFd, and MTF are seen to clearly improve in the order of their progression over 10 training epochs. The CNN with GAFd encoding achieved the highest performance: F1-score 78.98%, precision 92.89%, recall 81.53%, and validation accuracy 81.53%. GAFd captures temporal dynamics well, enabling the model to better differentiate patterns.

As a comparison, the MTF-based model was also good, with a final accuracy of 79.02% and a commendable F1 score of 77.49%, suggesting that MTF encoding is able to give meaningful state transition representations. The model trained with GAF came second, resulting in a final accuracy of 77.65% along with 74.79% as F1 score, although still showing steady improvement with epochs. Overall, despite the fact that all three encodings assisted the CNNs in learning discriminative features from time series data, GAFd encoding proved to be the most effective transformation in

this research, which delivered the best classification performance in all the evaluation metrics.

Tables 9 and 10 summarize the performance of CNN models trained on the WESAD dataset using different time-series encodings under time-axis and feature-axis sliding window strategies, respectively. When sliding windows are applied along the time axis (Table 9), GAFd achieves the best overall performance, with an F1-score of 78.98% and balanced precision and recall, indicating that derivative-based angular representations effectively capture short-term temporal dynamics associated with stress responses. MTF performs competitively, suggesting that transition-based representations also encode discriminative temporal structure, while GAFs show comparatively lower performance, reflecting limited sensitivity to rapid signal changes under this setting.

In contrast, when sliding windows are applied along the feature axis (Table 10), all encodings exhibit substantially higher performance, with accuracy and F1-scores exceeding 98%. MTF achieves peak accuracy (99.57%) and F1-score (99.54%), while GAFd converges faster with similar performance, reflecting its effectiveness in capturing dynamic interactions. Feature-axis windowing reduces temporal variability, allowing CNNs to capture stable inter-feature relationships and produce discriminative representations. The comparison indicates a trade-off: GAFd captures fast temporal changes, whereas MTF excels in stable feature interactions, with each suitable for different task requirements.

5.3 GRAD-CAM

The mid-sequence points are the primary contributors to class 3 prediction as shown in Fig. 4. Convolutional feature maps highlight these areas, and the Grad-CAM overlay verifies that the model attention is concentrated in the vicinity of rows/columns 3–5. This indicates that local contrast and angular diversity in the time series are important discriminative features for CNN-based time series classification using GAF representations.

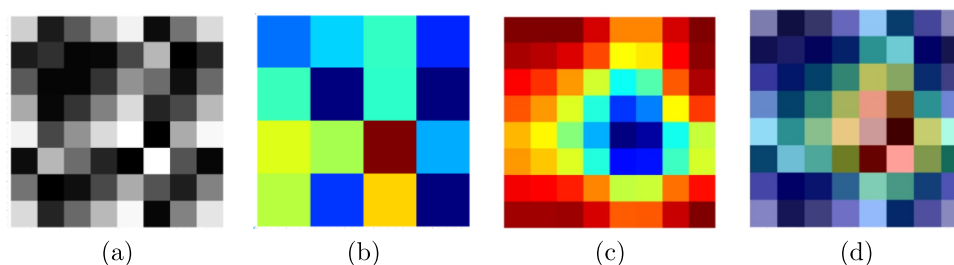


Fig. 4 Grad-CAM-based interpretability analysis for Class 3 (Physical Stress). From left to right: (a) the original GAF-encoded input image derived from the normalized physiological time series; (b) the corresponding feature map from the first convolutional layer, highlighting learned spatial-temporal patterns; (c) the Grad-CAM heatmap, where

warmer colors (red/yellow) indicate regions with higher contribution to the model's decision; and (d) the Grad-CAM overlay on the original GAF image, showing that the CNN focuses on high angular-contrast regions corresponding to abrupt temporal changes in the physiological signal

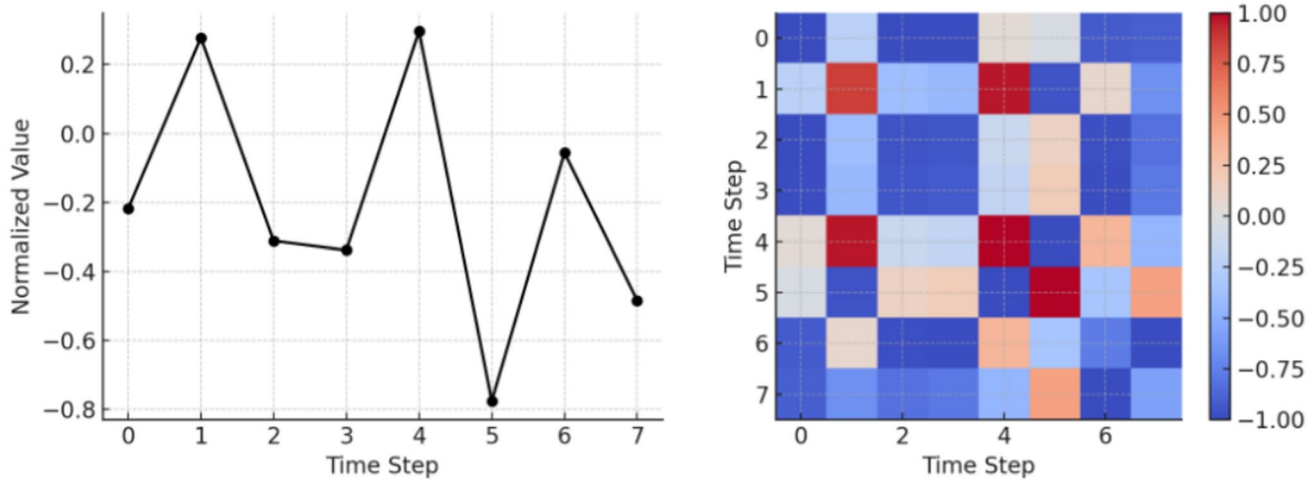


Fig. 5 Illustration of time-series to GAF transformation. (Left) Normalized physiological feature time series over eight time steps, where values are scaled to the range $[-1, 1]$ prior to encoding. (Right) Corresponding Gramian Angular Field (GAF) matrix constructed from the normalized series, in which both axes represent time steps and each pixel encodes the cosine of the angular difference between pairs of

time points. Warmer and cooler colors indicate high positive and negative angular similarity, respectively. This representation preserves temporal dependencies and highlights abrupt signal transitions as high-contrast regions, which are subsequently exploited by the CNN for discriminative feature learning

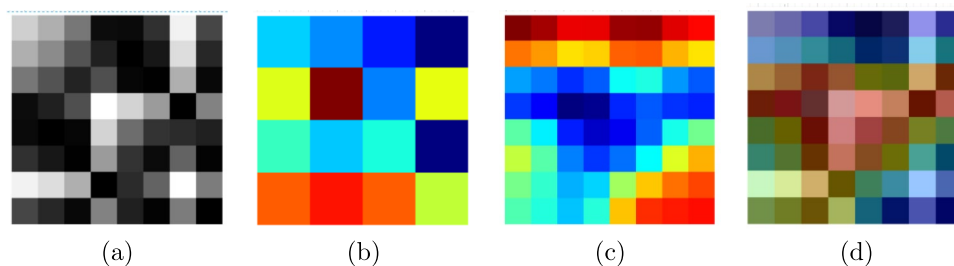


Fig. 6 Grad-CAM-based interpretability analysis for Class 2 (Emotional Stress). From left to right: (a) the GAF-encoded input image generated from the normalized physiological time series, (b) the feature map extracted from the first convolutional layer of the CNN highlighting learned spatial-temporal patterns, (c) the Grad-CAM heatmap indicating regions with the highest contribution to the model's predic-

tion (red/yellow = high importance, blue = low importance), and (d) the Grad-CAM overlay on the original GAF image. The highlighted high-activation regions correspond to areas of strong angular contrast in the GAF matrix, reflecting abrupt temporal changes in the underlying physiological signals that are characteristic of emotional stress responses

The resulting GAF matrix from the normalized feature as shown in Fig. 5 (right) series retains angular cosine-based pair-wise temporal relationships. The Grad-CAM heatmap identifies hotspots of high activation within the central GAF area at time steps 3–5. These locations have abrupt contrast - a dip-peak-dip shape - shown in Fig. 5 (left) that causes drastic angular changes and extreme GAF values (e.g., ± 1), reflecting dramatic temporal changes.

The Grad-CAM examination of Label 2 as shown in Fig. 6, Row 0 of the GAF matrix demonstrates that the CNN classifier is centered on angular high-contrast regions, especially GAF[3,6], GAF[3,7], GAF[6,3], and GAF[7,3] entries. The patterns can be attributed to interactions between highly extreme normalized values occurring at step 3 and steps 6–7. Extreme angular differences between minima and maxima results in elevated cosine responses,

that enable the extraction of elementary discriminative patterns in the GAF encoding.

Between steps 2 and 7, the signal rises noticeably (Fig. 7). In the GAF, this section aligns into a structured pattern towards the lower-right quadrant. Grad-CAM shows localized activation around rows and columns 6–7, where angular changes and contrast are most prominent. These attributes play a key role in directing the CNN towards a correct Class 2 prediction. Among several participants, Grad-CAM frequently focuses on angular changes between rows and columns 3–7, coupled with regions of strong contrast, indicating that the learned patterns are independent of individual subjects.

The Grad-CAM maps reveal that the CNN emphasizes on abrupt angular variations in the GAFs, especially the dip-peak-dip pattern around steps 3–5 and the steep increase near steps 6–7. Such swift angular shifts are consistent with

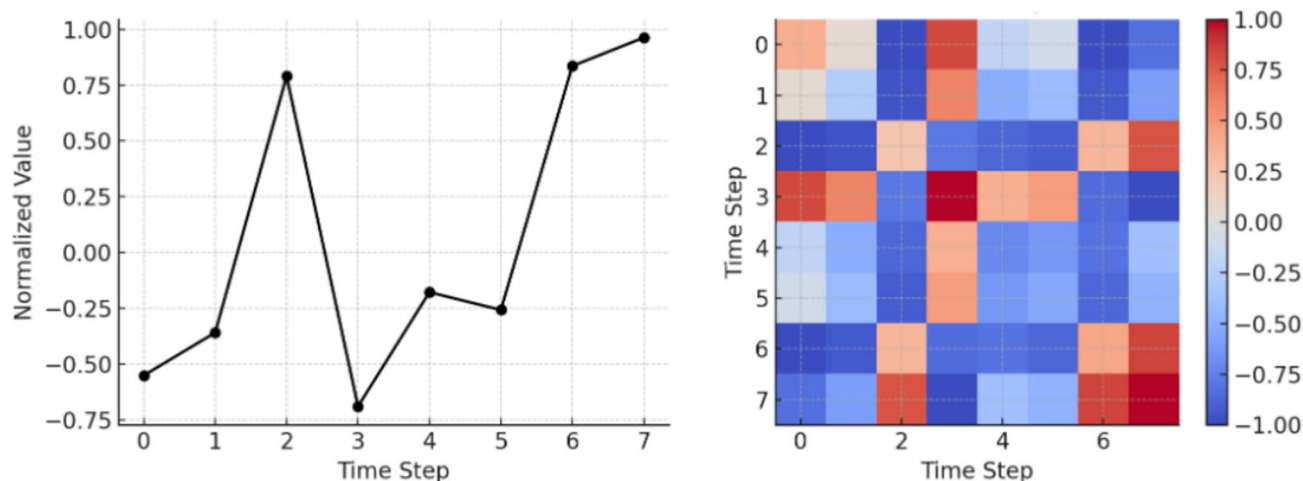


Fig. 7 Example of physiological time-series encoding using Gramian Angular Fields (GAF). (Left) Normalized physiological feature sequence over eight time steps, scaled to the range $[-1, 1]$, showing a pronounced upward trend with intermediate fluctuations. (Right) Corresponding GAF matrix representation, where both axes denote time steps and each pixel encodes the cosine similarity between angularly

mapped time points. Regions with strong positive (red) and negative (blue) values indicate high angular contrast, capturing temporal transitions and trends in the original signal. These structured patterns preserve temporal dependencies and form the basis for CNN-based discrimination of stress-related physiological dynamics

temporary EDA surges and concurrent reductions in HRV, both commonly linked with sympathetic activation during stress. The mid-sequence contrasts highlighted by Grad-CAM may therefore illustrate transient increases in electrodermal response or brief reductions in parasympathetic activity during acute stress episodes. In Class 2 examples, activation in the bottom-right quadrant corresponds to a more gradual increasing trajectory, which may correspond to periods of recovery or adaptation showing greater stability in EDA and HRV measures.

In the GAF representation, both axes represent sequential time points, while the color scale (-1 to $+1$) reflects the cosine similarity. Here higher absolute values indicate stronger angular contrast. For Classes 2 and 3, the Grad-CAM visualizations follow a blue-to-red scale to indicate regions that strongly influence the CNN's predictions. Together, these elements make it easier to relate timing, activation intensity, and angular variation to the core physiological signals shown in each figure.

6 Limitations

Although the proposed approach shows promising results, there are a few clear limitations to it. The NEURO and WESAD datasets cover only a limited range of demographics. This limitation reduces the broader applicability of the results. The effect is particularly relevant given individual differences in stress responses. Collecting data in controlled laboratory settings might not reflect the full range of real-world variability, including motion artifacts

and environmental noise. The high performance seen under cross-validation may partly reflect subject-specific patterns. However, applying LOSO evaluation helps reduce this effect. The relatively small size of the datasets limits the statistical power for comparing different models and encoding strategies. While Grad-CAM offers qualitative insights into model behavior, it has not been quantitatively validated or tested statistically against physiological ground truth.

7 Conclusion

Using encoded physiological signals along with CNNs leads to clear gains in the performance of stress detection systems. Converting physiological time-series into GAF, GAFd, and MTF image representations enable CNNs to better encode spatial structure, resulting in improved classification performance. GAF-based illustrations delivered substantial accuracy and showed uniform generalization across the WESAD and NEURO datasets. Grad-CAM analysis offered insight into physiological patterns associated with stress and clarified how the model arrived at its classifications. Our work demonstrates the benefits of integrating signal transformation with interpretable deep learning and supports effective and transparent stress detection for mental health applications.

8 Future works

Future work should involve larger and more diverse participant groups, by including a more diverse range of ages, genders, and ethnicities to enhance model robustness. Evaluating the method in operational environments will provide critical insights for assessing its robustness to noise and changing contextual conditions.

Integrating physiological data with facial and speech cues could offer deeper insight into how emotions are expressed. Such studies could capture the progression of stress responses in individuals and may help identify early indicators of chronic stress. Approaches such as subject-specific adaptation, quantitative explainability, and real-time deployment on wearable devices could make stress monitoring systems more personalized, interpretable, and practical.

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Data Availability Access the NEURO Dataset at: <https://physionet.org/content/noneeg/1.0.0/> Access the WESAD Dataset at: <https://ubi29.informatik.uni-siegen.de/usi/data/wesad.html>

Declarations

Competing interests The authors have no relevant conflicts of interest to disclose.

Research Involving Human and/or Animals Not Applicable.

Informed Consent Not Applicable.

References

- Al'Absi M (2006) Stress and Addiction: Biological and Psychological Mechanisms, 1st edn. Academic Press, Cambridge
- Birjandtalab J, Cogan D, Pouyan MB, Nourani M (2016) A non-eeg biosignals dataset for assessment and visualization of neurological status. In: 2016 IEEE International Workshop on Signal Processing Systems (SiPS), pp. 110–114. IEEE
- Cohen S, Janicki-Deverts D, Miller G (2007) Psychological stress and disease. *JAMA* 298:1685–1687
- De Santos Sierra A, Sanchez-Avila C, Guerra-Casanova J, Pozo G (2011) A stress-detection system based on physiological signals and fuzzy logic. *IEEE Trans Ind Electron* 58:4857–4865
- Giannakakis G, Grigoriadis D, Giannakaki K, Simantiraki O, Roniotis A, Tsiknakis M (2022) Review on psychological stress detection using biosignals. *IEEE Trans Affect Comput* 13:440–460. <https://doi.org/10.1109/TAFPC.2019.2927337>
- Gjoreski M, Luštrek M, Gams M, Gjoreski H (2017) Monitoring stress with a wrist device using context. *J Biomed Inform* 73:159–170. <https://doi.org/10.1016/j.jbi.2017.08.006>
- Heracleides A (2012) Work stress, obesity and the risk of type 2 diabetes: gender specific bidirectional effect in the whitehall ii study. *Obesity* 20:428–433
- Karthikeyan P, Murugappan M, Yaacob S (2013) Detection of human stress using short-term ecg and hrv signals. *J Mech Med Biol*. <https://doi.org/10.1142/s0219519413500383>
- Kyriakou K, Resch B, Sagl G, Petutschnig A, Werner C, Niederseer D, Liedgruber M, Wilhelm F, Osborne T, Pykett J (2019) Detecting moments of stress from measurements of wearable physiological sensors. *Sensors (Basel)* 19(17):3805. <https://doi.org/10.3390/s19173805>
- Li R, Liu Z (2020) Stress detection using deep neural networks. *BMC Med Inform Decis Mak* 20:1–10
- Lundberg U, Kadefors R, Melin B, Palmerud G, Hassmen P, Engstrom M, Dohns I (1994) Psychophysiological stress and emg activity of the trapezius muscle. *Int J Behav Med* 1:354–370
- Mirsamadi S, Barsoum E, Zhang C (2017) Automatic speech emotion recognition using recurrent neural networks with local attention. In: Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), pp. 2017–2021. IEEE, New Orleans, USA
- Mitra A (2008) Diabetes and stress: a review. *Stud Ethno-Med* 2:131–135
- Nandan AM, Choudhary D, Godbole I, et al. (2024) Classifying emotional states through eeg-derived spectrograms. In: 2024 IEEE 18th International Conference on Automatic Face and Gesture Recognition (FG), pp. 1–5. IEEE
- Oskooei A, Mai Chau S, Weiss J, Sridhar A, Rodríguez Martínez M, Michel B (2019) Destress: Deep learning for unsupervised identification of mental stress in firefighters from heart-rate variability (hrv) data. arXiv preprint [arXiv:1911.13213](https://arxiv.org/abs/1911.13213)
- Quadrini M, Capuccio A, Falcone D, Daberdaku S, Blanda A, Belanova L, Gerard G (2024) Stress detection with encoding physiological signals and convolutional neural network. *Mach Learn* 113(8):5655–5683
- Rastgoo MN, Nakisa B, Maire F, Rakotonirainy A, Chandran V (2019) Automatic driver stress level classification using multimodal deep learning. *Expert Syst Appl* 138:112793
- Schmidt P, Reiss A, Duerichen R, Marberger C, Van Laerhoven K (2018) Introducing wesad, a multimodal dataset for wearable stress and affect detection. In: Proceedings of the 20th ACM International Conference on Multimodal Interaction, pp. 400–408
- Seegerstrom S, Miller G (2004) Psychological stress and the human immune system: a meta-analytic study of 30 years of inquiry. *Psychol Bull* 130:601–630
- Setz C, Arnrich B, Schumm J, La Marca R, Tröster G, Ehlert U (2010) Discriminating stress from cognitive load using a wearable eda device. *IEEE Trans Inf Technol Biomed* 14:410–417
- Shikha S, Sethia D, Indu S (2024) Optimization of wearable biosensor data for stress classification using machine learning and explainable ai. *IEEE Access* 12:169310–169327
- Sioni R, Chittaro L (2015) Stress detection using physiological sensors. *IEEE Comput* 48:26–33
- Smets E, Rios-Velazquez E, Schiavone G, Chakroun I, D'Hondt E, De Raedt W, Cornelis J, Janssens O, Van Hoecke S, Claes S et al (2018) Large-scale wearable data reveal digital phenotypes for daily-life stress detection. *npj Digital Medicine* 1(1):67. <https://doi.org/10.1038/s41746-018-0074-9>
- Souza A, Melchiades MB, Rigo SJ, Ramos GdO (2022) Mostress: A sequence model for stress classification. In: 2022 International Joint Conference on Neural Networks (IJCNN), pp. 1–8. IEEE
- Step toe A, Kivimaki M (2013) Stress and cardiovascular disease: an update on current knowledge. *Annu Rev Public Health* 34:337–354
- The American Institute of Stress: Daily Life. Accessed: 2020-02-29 (2020). <https://www.stress.org/daily-life>
- Tzirakis P, Trigeorgis G, Nicolaou M, Schuller B, Zafeiriou S (2017) End-to-end multimodal emotion recognition using deep neural networks. *IEEE J Sel Top Signal Process* 11:1301–1309

- Van Praag H (2004) Can stress cause depression? *Prog Neuropsychopharmacol Biol Psychiatry* 28:891–907
- Wang J, Lin C, Yang Y (2013) A k-nearest-neighbor classifier with heart rate variability feature-based transformation algorithm for driving stress recognition. *Neurocomputing* 116:136–143
- Wang S, Li Y, Zhang T, Li K (2024) Tagformer: a multimodal physiological signals fusion network for pilot stress recognition. *IEEE Sens J* 24(13):20842–20854

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